DeVan- PNS Pathology — Practice Test

1. A 45-year-old man presents with progressive distal symmetric sensory loss beginning in the feet and ascending proximally over several months. Electrophysiology shows reduction in compound muscle action potential amplitude with preserved conduction velocity. Which pathologic process best explains these findings?

A. Demyelination of Schwann cells
B. Axonal degeneration with loss of nerve fibers
C. Compression of nerve fascicles
D. Inflammation of dorsal root ganglia

> **Explanation: Explanation:**
> The clinical presentation of distal symmetric sensory loss ascending proximally is characteristic of a polyneuropathy with axonal degeneration (length-dependent). The electrophysiologic hallmark of axonal neuropathy is reduction in amplitude (reflecting loss of functioning axons) with relatively preserved conduction velocity. Demyelinating neuropathies present with slowed conduction velocity. The ascending pattern helps distinguish from neuronopathies which can affect proximal and distal regions.

2. A 28-year-old woman develops acute onset weakness and hyporeflexia that progresses over several days, eventually requiring mechanical ventilation. CSF analysis shows elevated protein with normal glucose. Which mechanism best explains the neurologic findings?

A. Acute axonal injury from toxic exposure
B. Acute demyelination of peripheral nerves
C. Compression of nerve roots
D. Destruction of dorsal root ganglia neurons

> **Explanation: Explanation:**
> This clinical presentation is consistent with Guillain-Barré Syndrome (Acute Inflammatory Demyelinating Polyneuropathy). The rapid ascending paralysis, elevated CSF protein with normal glucose, and areflexia are classic findings. AIDP is characterized by acute demyelination of peripheral nerves with immune-mediated attack on Schwann cells, not primary axonal injury. Neuronopathies would show proximal and distal involvement patterns different from this ascending paralysis.

3. A 52-year-old man with poorly controlled diabetes for 15 years develops distal sensory loss in a stocking-glove distribution, with particular involvement of pain and temperature sensation. Which nerve fiber type is primarily affected?

A. Large myelinated A-alpha fibers mediating motor function
B. Large myelinated A-beta fibers mediating touch
C. Small unmyelinated C fibers and small A-delta fibers conveying pain and temperature
D. Preganglionic sympathetic fibers

> **Explanation: Explanation:**
> Diabetic neuropathy initially affects small unmyelinated C fibers and small A-delta fibers which convey pain and temperature sensation, causing early loss of pain and temperature sense in a distal symmetric pattern. Large diameter myelinated fibers that transmit touch and motor signals are affected later. Hyperglycemia-induced oxidative stress triggers neuronal apoptosis. The stocking-glove distribution indicates a classic distal symmetric sensorimotor polyneuropathy.

4. A 35-year-old woman presents with progressive weakness and fatigability that worsens with sustained muscle activity and improves with rest. She has ptosis and diplopia. Antibodies against acetylcholine post-synaptic receptors are detected. Which of the following best explains the pathophysiology?

A. Blockade of acetylcholine receptors by autoantibodies preventing muscle contraction
B. Inhibition of voltage-gated calcium channels preventing acetylcholine release
C. Congenital loss-of-function mutations in acetylcholine receptor clustering
D. Clostridial toxin blocking acetylcholine release from presynaptic terminals

> **Explanation: Explanation:**
> This patient has Myasthenia Gravis, an autoimmune disorder with autoantibodies against acetylcholine post-synaptic receptors. These antibodies block receptor function and prevent muscle contraction, causing weakness that worsens with exertion. Option B describes Lambert-Eaton Myasthenic Syndrome (LEMS). Option C describes congenital myasthenic syndromes. Option D describes botulism. Myasthenia Gravis is strongly associated with thymic abnormalities including thymoma and thymic hyperplasia.

5. A 58-year-old man with small cell lung carcinoma develops proximal muscle weakness and fatigue. EMG shows incremental increase in compound muscle action potential with repetitive stimulation. Antibodies against presynaptic voltage-gated calcium channels are found. What is the most likely diagnosis?

A. Myasthenia Gravis
B. Lambert-Eaton Myasthenic Syndrome
C. Botulism
D. Congenital myasthenic syndrome

> **Explanation: Explanation:**
> Lambert-Eaton Myasthenic Syndrome (LEMS) is characterized by autoantibodies against presynaptic voltage-gated calcium channels, which inhibit acetylcholine release. The key distinguishing feature is that repetitive muscle stimulation IMPROVES the response (incremental increase), unlike myasthenia gravis where repetitive stimulation causes decremental response. About half of LEMS patients have underlying malignancy, classically small cell lung carcinoma. The proximal weakness pattern is typical.

6. A 32-year-old man from an endemic area develops demyelinating and remyelinating segments within peripheral nerves with shortening of internodes and thinner myelin sheaths. Acid-fast bacilli are identified within Schwann cells. Which infectious organism is responsible?

A. Mycobacterium tuberculosis
B. Mycobacterium leprae
C. Treponema pallidum
D. Borrelia burgdorferi

> **Explanation: Explanation:**
> This patient has lepromatous (multibacillary) leprosy caused by Mycobacterium leprae. The organism invades Schwann cells directly, causing segmental demyelination and remyelination with loss of axons. The resulting shortened and thin internodes are characteristic. Pain fibers are particularly affected, leading to numbness and unknown injuries with large ulcers. Tuberculoid leprosy causes localized granulomatous responses with dermal nodules and more localized nerve involvement.

7. A 48-year-old woman presents with mononeuritis multiplex affecting random individual peripheral nerves. Laboratory workup shows elevated inflammatory markers and positive MPO-ANCA. Which pathologic mechanism best explains the peripheral nerve involvement?

A. Direct bacterial invasion of nerve sheaths
B. Non-infectious inflammation of blood vessels supplying peripheral nerves
C. Demyelination from circulating anti-myelin antibodies
D. Toxic degeneration from chemotherapeutic agents

> **Explanation: Explanation:**
> Mononeuritis multiplex with MPO-ANCA positivity indicates vasculitis-associated neuropathy, likely microscopic polyangiitis or eosinophilic granulomatosis with polyangiitis (Churg-Strauss). The mechanism involves non-infectious inflammation of blood vessels supplying peripheral nerves, leading to ischemic injury and random individual nerve involvement (not symmetric like polyneuropathies). MPO-ANCA-associated vasculitides are the most common vasculitis-associated neuropathies, more so than PR3-ANCA-associated disease.

8. A 28-year-old construction worker sustained a complete laceration of his median nerve in the forearm. Pathologic examination of the distal nerve segment shows loose myelin layers and lipid droplets within nerve fibers. Which process is occurring?

A. Wallerian degeneration
B. Onion bulb formation
C. Schwann cell proliferation with remyelination
D. Axonal regeneration with increased diameter

> **Explanation: Explanation:**
> When an axon is transected, the distal axon segment undergoes Wallerian degeneration. The pathologic findings of loose myelin layers and lipid droplets reflect the breakdown of the myelin sheath and axonal debris. This is a normal consequence of axonal injury distal to the site of transection. The proximal segment attempts regeneration, but the distal segment must degenerate before any possibility of reinnervation. Onion bulb formation results from recurrent demyelination and remyelination, not acute axonal injury.

9. A 55-year-old woman with a 20-year history of rheumatoid arthritis develops peripheral neuropathy with recurrent demyelination and remyelination creating 'onion bulb' formations on nerve biopsy. Which histologic finding would you expect to see?

A. Multiple concentric layers of Schwann cell processes around a single axon
B. Complete loss of myelin with exposed axons
C. Infiltration of nerve by acid-fast bacilli
D. Vascular fibrinoid necrosis within nerve fascicles

> **Explanation: Explanation:**
> Onion bulb formations result from chronic recurrent demyelination and remyelination as seen in chronic inflammatory demyelinating poly(radiculo)neuropathy (CIDP) and in systemic inflammatory diseases like rheumatoid arthritis, Sjogren syndrome, and lupus. The characteristic pathologic finding is multiple concentric layers of Schwann cell processes surrounding a single axon, creating a characteristic 'onion bulb' appearance. This is distinct from acute demyelination or infectious processes.

10. A 62-year-old man develops progressive distal sensorimotor neuropathy after receiving cisplatin chemotherapy for testicular cancer. Which nerve components are primarily damaged?

A. Schwann cells and myelin sheaths
B. Axons of peripheral neurons
C. Blood vessels supplying the nerves
D. Dorsal root ganglia neurons

> **Explanation: Explanation:**
> Cisplatin causes a dose-dependent axonal neuropathy (axonal degeneration), not demyelination. It is one of the chemotherapeutic agents that produce toxic peripheral neuropathy. Other chemotherapeutics with neuropathic effects include vinca alkaloids (vincristine, vinblastine) and taxanes. Platinum compounds cause neurotoxicity through oxidative stress and direct axonal injury. The neuropathy typically presents as distal sensorimotor symptoms and can be dose-limiting.

11. A 4-year-old child presents with developmental delay, intellectual disability, and multiple cutaneous lesions including café-au-lait spots and Lisch nodules. MRI shows neurofibromas and optic nerve gliomas. Which gene mutation is responsible for this syndrome?

A. NF2 gene on chromosome 22
B. NF1 gene on chromosome 17
C. PMP22 gene on chromosome 17
D. Transthyretin gene

> **Explanation: Explanation:**
> This child has Neurofibromatosis Type 1 (NF1), caused by loss-of-function mutations in the NF1 gene on chromosome 17q11.2. NF1 codes for neurofibromin, a tumor suppressor with GTPase activity. Without functional neurofibromin, RAS remains trapped in an active state, leading to tumorigenesis. NF1 occurs in approximately 1:3000 people and is associated with neurofibromas, MPNSTs, optic nerve gliomas, pheochromocytomas, intellectual disability, seizures, skeletal defects, Lisch nodules, and café-au-lait spots.

12. A 45-year-old man presents with bilateral eighth cranial nerve schwannomas causing hearing loss, and multiple meningiomas. Which of the following best describes the genetic basis of his condition?

A. Loss of neurofibromin function causing RAS hyperactivation
B. Loss of merlin function affecting cytoskeletal regulation and cell adhesion
C. Duplication of PMP22 gene causing myelin instability
D. Transthyretin mutations causing amyloid accumulation in nerves

> **Explanation: Explanation:**
> This patient has Neurofibromatosis Type 2 (NF2), characterized by bilateral eighth nerve schwannomas and multiple meningiomas. NF2 is caused by mutations in the NF2 gene on chromosome 22q12, which codes for merlin, a cytoskeletal protein that regulates signal pathways and controls cell shape, growth, and adhesion. NF2 is much rarer (1:40,000-50,000) than NF1. Loss of merlin function differs mechanistically from NF1's effects on RAS signaling.

13. A 38-year-old woman presents with progressive weakness associated with a malignant plasma cell clone secreting monoclonal immunoglobulin. She has organomegaly, endocrinopathy, skin changes, and demyelinating neuropathy. What is this syndrome called?

A. Paraneoplastic sensorimotor neuropathy
B. POEMS syndrome
C. Lambert-Eaton Myasthenic Syndrome
D. Amyloidosis with familial polyneuropathy

> **Explanation: Explanation:**
> POEMS syndrome (Polyneuropathy, Organomegaly, Endocrinopathy, Monoclonal gammopathy, Skin changes) is associated with demyelinating neuropathy and a monoclonal gammopathy. The neoplastic B cells secrete IgG or IgG fragments that cause the peripheral nerve dysfunction. This is distinct from paraneoplastic sensorimotor neuropathy associated with lung cancer (particularly small cell carcinoma with CD8+ cytotoxic T cell attack on dorsal root ganglia) and from LEMS. The combination of demyelinating neuropathy with systemic manifestations and monoclonal protein is pathognomonic.

14. A 52-year-old woman develops carpal tunnel syndrome with progressive median nerve compression within the space created by the transverse carpal ligament. Which of the following is NOT a known risk factor for her condition?

A. Inflammatory arthritis affecting the wrist
B. Pregnancy-related fluid retention
C. Hypothyroidism
D. Recent viral infection causing demyelination

> **Explanation: Explanation:**
> Carpal tunnel syndrome results from compression of the median nerve within the carpal tunnel space. Known risk factors include inflammatory arthritis, pregnancy (from fluid retention), hypothyroidism, amyloidosis, and diabetes. Recent viral infection causing demyelination is not a recognized risk factor for carpal tunnel syndrome. Compression neuropathies result from mechanical forces on nerves, not from infectious or inflammatory demyelinating processes affecting the nerve itself.

15. A 55-year-old man presents with progressive weakness and a palpable mass along a large peripheral nerve in the thigh. Biopsy shows a highly cellular spindle cell tumor with fascicular architecture and mitotic figures. Immunohistochemistry shows strong S100 staining but negative neurofibromin staining. What is the most likely diagnosis?

A. Schwannoma of the thigh
B. Benign neurofibroma
C. Malignant Peripheral Nerve Sheath Tumor (MPNST)
D. Plexiform neurofibroma

> **Explanation: Explanation:**
> The clinical and pathologic features point to MPNST: large tumor along a major nerve with high cellularity, mitotic figures, and fascicular architecture. The negative neurofibromin staining indicates loss of NF1 gene product. MPNSTs are high-grade malignancies (85% are high-grade) and are associated with large peripheral nerves. About 50% of MPNSTs arise in patients with NF1, often from malignant transformation of plexiform neurofibromas. Schwannomas are benign and encapsulated. S100 positivity is seen in both benign and malignant nerve sheath tumors but the high cellularity and mitotic activity indicate malignancy.